

Number System - MCQs (Hard Level)

Campus Placement Level | Quantitative Aptitude | 30 Questions with Correct Answers

Question 1

What are the last two digits of 7^{2023} ?

- A) 07
- B) 43
- C) 49
- D) 93

Correct Answer: B) 43

Question 2

What is the remainder when 3^{200} is divided by 13?

- A) 1
- B) 3
- C) 9
- D) 12

Correct Answer: C) 9

Question 3

What is the remainder when 17^{2025} is divided by 18?

- A) 17
- B) 1
- C) 0
- D) 16

Correct Answer: A) 17

Question 4

How many trailing zeros are there in $250!$?

- A) 50
- B) 62
- C) 60
- D) 64

Correct Answer: B) 62

Question 5

What is the highest power of 3 that divides $100!$?

- A) 33
- B) 46
- C) 50
- D) 48

Correct Answer: D) 48

Question 6

How many positive divisors does 2520 have?

- A) 36
- B) 42
- C) 48
- D) 54

Correct Answer: C) 48

Question 7

What is the sum of all positive divisors of 360?

- A) 1170
- B) 1080
- C) 1254
- D) 1008

Correct Answer: A) 1170

Question 8

How many positive divisors of 3600 are perfect squares?

- A) 8
- B) 10
- C) 15
- D) 12

Correct Answer: D) 12

Question 9

What is the remainder when $50!$ is divided by 53? (53 is prime)

- A) 1
- B) 27
- C) 26
- D) 52

Correct Answer: C) 26

Question 10

What is the value of Euler's totient function $\phi(360)$, i.e. how many integers from 1 to 360 are coprime to 360?

- A) 72
- B) 96
- C) 90
- D) 120

Correct Answer: B) 96

Question 11

How many integers from 1 to 100 are coprime to 100?

- A) 40
- B) 20
- C) 30
- D) 50

Correct Answer: A) 40

Question 12

What is the smallest positive number which leaves remainders 5, 9 and 13 when divided by 8, 12 and 16 respectively?

- A) 43
- B) 45
- C) 47
- D) 93

Correct Answer: B) 45

Question 13

What is the HCF of $2^{48} - 1$ and $2^{36} - 1$?

- A) 4095
- B) 63
- C) 1023
- D) 65535

Correct Answer: A) 4095

Question 14

What is the remainder when 2^{2023} is divided by 15?

- A) 2
- B) 4
- C) 1
- D) 8

Correct Answer: D) 8

Question 15

What is the decimal value of the binary number 1011.101?

- A) 11.5
- B) 11.125
- C) 11.625
- D) 13.625

Correct Answer: C) 11.625

Question 16

How many digits does 2^{100} have? (Use $\log_{10} 2 = 0.30103$)

- A) 30
- B) 31
- C) 32
- D) 33

Correct Answer: B) 31

Question 17

How many trailing zeros does $100!$ have when written in base 12?

- A) 48
- B) 24
- C) 47
- D) 97

Correct Answer: A) 48

Question 18

What is the sum of the digits of the number $10^{20} - 7$?

- A) 171
- B) 177
- C) 180
- D) 174

Correct Answer: D) 174

Question 19

What is the value of $1^2 + 2^2 + 3^2 + \dots + 20^2$?

- A) 2870
- B) 2470
- C) 2850
- D) 3080

Correct Answer: A) 2870

Question 20

How many integers from 1 to 1000 are divisible by exactly one of 3 and 5 (but not both)?

- A) 467
- B) 534
- C) 401
- D) 333

Correct Answer: C) 401

Question 21

How many integers from 1 to 500 are divisible by none of 2, 3 and 5?

- A) 100
- B) 134
- C) 150
- D) 166

Correct Answer: B) 134

Question 22

In how many ways can 360 be written as a product of two coprime factors $a \times b$ with $a < b$?

- A) 4
- B) 2
- C) 8
- D) 12

Correct Answer: A) 4

Question 23

What is the smallest positive number that leaves remainders 2, 3 and 4 when divided by 3, 5 and 7 respectively?

- A) 23
- B) 103
- C) 53
- D) 105

Correct Answer: C) 53

Question 24

What is the sum of all 3-digit numbers divisible by 7?

- A) 70336
- B) 70128
- C) 70350
- D) 71052

Correct Answer: A) 70336

Question 25

How many 3-digit numbers have the sum of their digits equal to 10?

- A) 45
- B) 54
- C) 63
- D) 66

Correct Answer: B) 54

Question 26

How many 4-digit numbers have all four digits distinct?

- A) 5040
- B) 3024
- C) 4500
- D) 4536

Correct Answer: D) 4536

Question 27

What is the unit digit of $1! + 2! + 3! + \dots + 50!$?

- A) 0
- B) 1
- C) 3
- D) 7

Correct Answer: C) 3

Question 28

What is the highest power of 2 that divides $100!$?

- A) 50
- B) 94
- C) 100
- D) 97

Correct Answer: D) 97

Question 29

What is the remainder when $1^3 + 2^3 + 3^3 + \dots + 10^3$ is divided by 7?

- A) 1
- B) 0
- C) 2
- D) 3

Correct Answer: A) 1

Question 30

If $N = 2^3 \times 3^4 \times 5^2$, how many positive divisors of N are multiples of 6?

- A) 36
- B) 24
- C) 45
- D) 60

Correct Answer: A) 36
